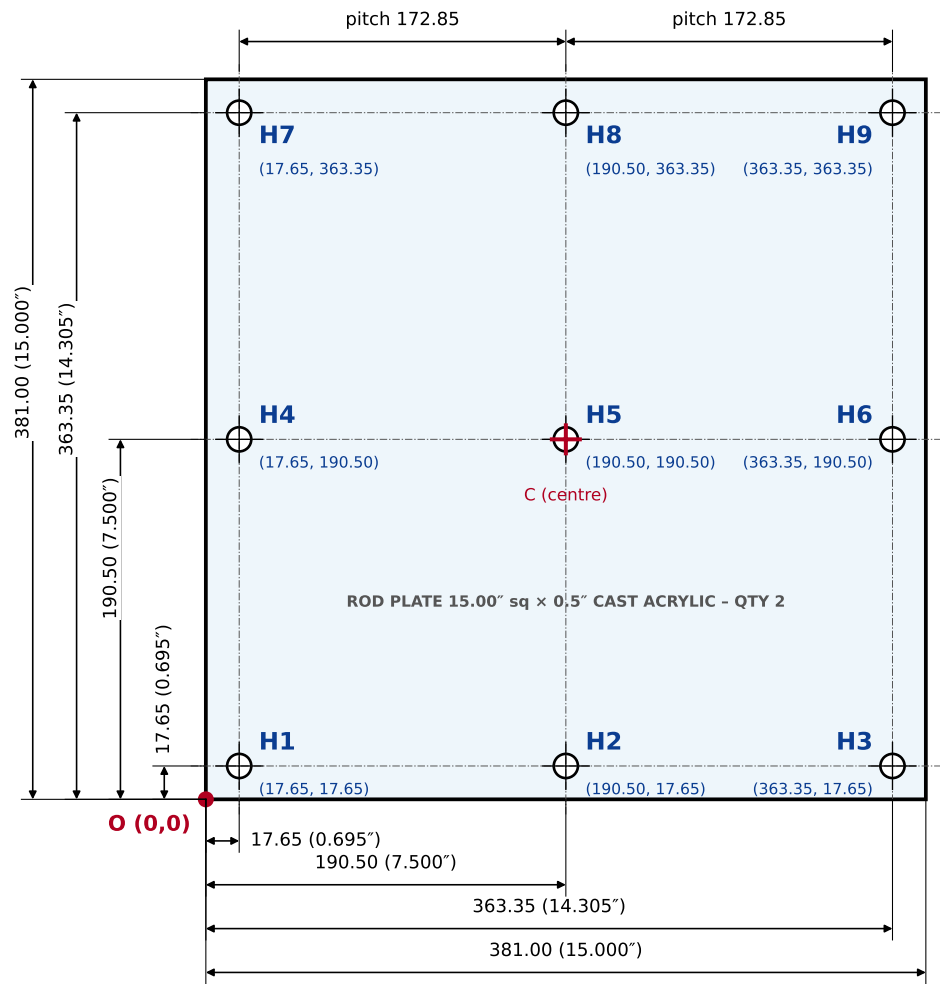


SHEET 1 - DRILL PATTERN, ROD PLATE (make 2 identical) - Exp B · 433.59 MHz

Top view, scale 1:4. 3 × 3 grid, pitch λ/4 = 172.85 mm. 9 holes Ø3.175 mm REAM THRU. Origin O = lower-left corner.



Hole symbols enlarged – NOT to scale. Work from the coordinates.

SLW HUB · ACRYLIC ROD-HOLDING JIG

Exp B · 433.59 MHz · FACT (M&W frequency and geometry)

DRAFT - for Dan's review

Experiment	Experiment B - M&W 433.59 MHz replication
Frequency	433.59 MHz
Wavelength	$\lambda = c/f = 691.42 \text{ mm}$
Rod length	346.00 ±1.0 mm (13.622")
	$\lambda/2 = 345.71 \text{ mm (free space)}$
Rod spacing	$\lambda/4 = 172.85 \text{ mm (6.8053")}$
Rod dia	1/8" = 3.175 mm (brass, then W1)
Hole	Ø3.175 mm REAM THRU, +0.013/-0
	size to MEASURED rod (see sheet 2)
Hole position	±0.25 mm (±0.010")
Material	CAST acrylic, 0.5" (12.70 mm)
Rod plate	15.00 × 15.00" (381.0 mm sq), QTY 2
Scale	TOP VIEW 1:4 on letter (print 100%)
Date / by	2026-09-28 · Grok Bot for Dan Britton
Sheet	1 of 3 · DRILL PATTERN (TOP VIEW)

Hole	X mm	Y mm	X in	Y in
H1	17.65	17.65	0.6947	0.6947
H2	190.50	17.65	7.5000	0.6947
H3	363.35	17.65	14.3053	0.6947
H4	17.65	190.50	0.6947	7.5000
H5	190.50	190.50	7.5000	7.5000
H6	363.35	190.50	14.3053	7.5000
H7	17.65	363.35	0.6947	14.3053
H8	190.50	363.35	7.5000	14.3053
H9	363.35	363.35	14.3053	14.3053

Origin O = lower-left corner. Hole Ø3.175 THRU.

Centre-origin table: sheet 2.

QTY 2 plates, identical drilling.

Position tol ± 0.25 mm (± 0.010 ").

SHEET 3 - CUT LIST, ASSEMBLY, NOTES - Exp B · 433.59 MHz

Part	Qty	Size (in)	Size (mm)	Operation
Rod plate	2	15 × 15"	381 × 381	drill/ream 9 holes (sheet 1)
H-array base	1	15 × 15"	381 × 381	dado 0.5" +0.004 wide × 0.25" deep, centred
V-array legs	2	15 × 7.625"	381 × 194	square ends, no holes
Fit-test coupon	1	3 × 3"	76 × 76	3 trial holes: reamer sizes

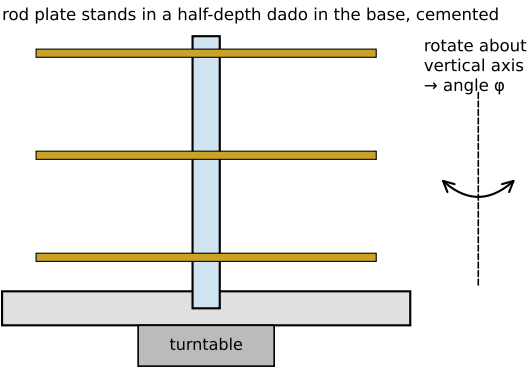
All parts CAST acrylic 0.5" (12.70 mm), clear. Net area 912.8 in²; +20 % margin 1095.3 in²; acrylic mass ≈ 8.90 kg.

BUY: 1 sheet 24 × 48" × 0.5" cast acrylic (fits by guillotine nesting, 1/8" kerf) - or order the parts cut to size.

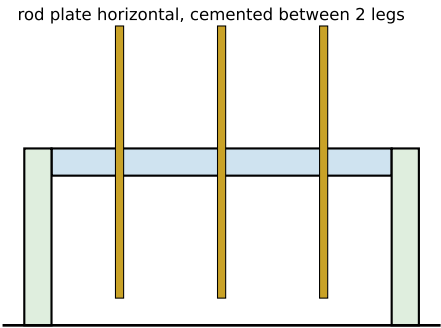
V-array leg height 7.625" = rod protrusion 166.7 mm + 25 mm clearance to the stand (IDEA).

Join: Weld-On 16 (or 4) solvent cement; anneal machined parts before cementing (acrylic makers' advice). NO metal fasteners near the rods.

H ARRAY (rods horizontal, rotatable)



V ARRAY (rods vertical, fixed)



Keep steel (turntable bearing, screws) ≥ λ/(2π) = 110 mm from the nearest rod: add a wood/plastic riser under the base if needed (IDEA).

Stand height: centre rod H5 on the TX-RX ball axis. Height is an OPEN item (depends on ball mounts).

- FACT** Monstein & Wesley, EPL 59(4) 514 (2002) p.516: 'cubic array of 9 wires of length λ/2 (34.6 cm)' as polarizer-analyzer at 433.59 MHz; brass rods; Fig. 2 photo: rods pass through ONE board at mid-length; 'Two such arrays with wires perpendicular to the direction of propagation and to each other'. Rod spacing, rod diameter and board size are NOT stated. Ball antennas 6 cm dia.
- FACT** Butterworth et al., J. Sci. Explor. 27(1) 13-23 (2013) p.16 (446 MHz rebuild): '3 × 3 arrays, a half-wavelength long and a quarter-wave apart, one horizontal and one vertical. The horizontal polarizer can be rotated.' → λ/4 pitch comes from Butterworth.
- FACT** Frequency/scaling: 433.59 MHz and 34.6 cm are M&W's own numbers.
- HYP** 'Cubic' read as: 3 × 3 at λ/4 spans λ/2, equal to the rod length, so the array fills a λ/2 cube.
- IDEA** Acrylic replaces the earlier HDPE fixtures (rod-fixtures rev A) as requested. Single plate at rod mid-length follows M&W Fig. 2.
- IDEA** Plate t/L = 0.037: plastic covers a small part of each rod. Rods electrically FLOATING, no wires, as in M&W.
- IDEA** 1/8" rods: stock on hand/ordered (brass + W1), same holes for both; half-rod self-weight sag 0.13 mm (brass).
- NOTE** Rod material enters M&W's model only through σ (eq. 7); geometry is identical for brass and W1 rods.

SLW HUB · ACRYLIC ROD-HOLDING JIG

Exp B · 433.59 MHz · FACT (M&W frequency and geometry)

DRAFT - for Dan's review

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Hole	Ø3.175 mm REAM THRU, +0.013/-0
	size to MEASURED rod (see sheet 2)
Hole position	±0.25 mm (±0.010")
Material	CAST acrylic, 0.5" (12.70 mm)
Rod plate	15.00 × 15.00" (381.0 mm sq), QTY 2
Scale	sketches NOT to scale
Date / by	2026-09-28 · Grok Bot for Dan Britton
Sheet	3 of 3 · CUT LIST, ASSEMBLY, NOTES

ASSEMBLY ORDER

- Cut & drill coupon; pick reamer that gives a snug push fit by hand.
- Drill/ream both rod plates together (stacked, pinned) so patterns match.
- Anneal, then cement H plate into base dado; cement V plate between legs.
- Push rods in, centre them (equal 166.7 mm each side), add O-rings.
- Brass first; later swap to W1 rods (same holes, same geometry).

Files: numbers.json, src/*.py
/workspace/sim-lab/rod-jig/